The Canadian Urban Environmental Health Research Consortium

Canue Metadata - Greenness AVHRR - NDVI

2026-11-06

DATA SET INFORMATION

Dataset Code: GRAVH_AMN_YY

Description:

The National Oceanic and Atmospheric Administration (NOAA) AVHRR series satellites were used to generate 1 km, NDVI composites based on surface reflectance data, annually producing 36 composites every 10-days for all of Canada from 1985 to 2013 (Latifovic et al., 2005). Resulting cloud free NDVI composites were used to derive NDVI metrics both annually and for the growing season (defined as May 1st through August 31st) for all 6-digit DMTI Spatial single link postal code locations in Canada.

Keywords: greenness|AVHRR|NDVI|satellite monitoring|normalized difference vegetation index|annual

Place Keywords: Canada|national

GEOSPATIAL REFERENCE

Upper Left Corner: 65.14N , -141.02W

Lower Right Corner: 41.68N , -52.62W

Coordinate System: GCS_WGS84 - EPSG:4326

Geometry Type: POINT - Units: Decimal Degree

Geometry Data Source: DMTI Spatial Inc. (postal codes)

QUALITY ASSESSMENT

QA/QC Procedures:

CANUE did not assess the quality of the AVHRR data. Users should review the documentation provided in the recommended citation, and in the supporting documentation.

Geographic Coordinate Positional Accuracy:

These metrics are linked to the corresponding annual postal codes files for mapping and analysis purposes. Refer to the postal code metadata file in Supporting Documentation for more information.

Vertical Positional Accuracy: N/A

Attribute Accuracy: N/A

Data Validity: NoData = -9999 (for numeric fields) - NoData=null (for category fields) - Data insufficient to calculate value = -1111

Associated Files: N/A

Data Comment:

N/A

DATA SOURCE

Data Source

AVHRR satellite product downloaded from NEODF, DMTI Spatial Inc. postal codes. See Supporting Documentation

Spatial Resolution: 1000 meters

Data Preparation Date: 2017-10-16

Beginning Date: 1985

End Date: 2013

Sampling Frequency of Data: Annual

Years Available:
1985 - 1986 - 1987 - 1988 - 1989 - 1990 - 1991 - 1992 - 1993 - 1994 - 1995 - 1996 - 1997 - 1998 - 1999 - 2000 - 2001 - 2002 - 2003 - 2004 - 2005 - 2006 - 2007 - 2008 - 2009 - 2010 - 2011 - 2012 - 2013

MAINTENANCE

Description: N/A

File Type: Comma separated values(.csv)

File Size: Individual year files range from 13 MB to 19 MB in size

Number of Data Files: 29

DATA USE CONDITIONS

The Data User is REQUIRED:
(i) to acknowledge data sources listed under Acknowledgement(s)
(ii) cite the publication(s) listed under Recommended Citation(s) as the providers and source of these data when using them in support of research, analysis, operations, policy decision or any other undertaking including publication
(iii) complete and sign the CANUE Data Use and Sharing Agreement (available at http://canue.ca/data/), in which the name and signature of the researcher/analyst who takes responsibility for ensuring all conditions are met.

Data Sharing Restrictions:
These data files are provided solely for the purposes stated in the CANUE Data Sharing and Use Agreement and should not be re-distributed for any reason. These data also contain proprietary postal code data and may only be used for the project named in the CANUE Data Sharing and Use Agreement. Data can be shared only within a project team for the exclusive purposes of teaching, academic research and publishing, and/or planning of educational services in accordance to DMTI End User Agreement associated with the Spatial Mapping Academic Research Tools (SMART) Program.

Include the following references in any publications resulting from the use of these data:
[1] Latifovic, R., Trishchenko, A., Chen, J., Park, W., Khlopenkov, K., Fernandes, R., Pouliot, D., Ungureanu, C., Luo, Y., Wang, S., Davidson, A., and Cihlar, J. 2005. Generating historical AVHRR 1-km baseline satellite data records over Canada suitable for climate change studies. Canadian Journal of Remote Sensing, vol. 31:324-34.
[2] CanMap Postal Code Suite v2015.3. [computer file] Markham: DMTI Spatial Inc., 2015.

Include the following acknowledgements:
1. NDVI metrics, indexed to DMTI Spatial Inc. postal codes , were provided by CANUE (Canadian Urban Environmental Health Research Consortium)

SUPPORT DOCUMENTATION

1 - AVHRR data via NEODF (https://neodf.nrcan.gc.ca/neodf_cat3/index.php?lang=en)
2 - Khlopenkov K. and Trishchenko A. 2006. SPARC: New Cloud, Snow, and Cloud Shadow Detection Scheme for historical 1-km AVHRR data over Canada. Journal of Atmospheric and Oceanic Technology, vol. 24. ()
3 - Latifovic, R., Trishchenko, A., Chen, J., Park, W., Khlopenkov, K., Fernandes, R., Pouliot, D., Ungureanu, C., Luo, Y., Wang, S., Davidson, A., and Cihlar, J. 2005. Generating historical AVHRR 1-km baseline satellite data records over Canada suitable for climate change studies. Canadian Journal of Remote Sensing, vol. 31:324-34. ()
4 - Postal Code metadata (https://canuedata.ca/docs/CANUE-Browser-Metadata-PostalCodes.pdf)

VARIABLES

GRAVHYY_01 - Annual maximum NDVI

Annual maximum NDVI (range -1 to 1)

GRAVHYY_02 - Annual average NDVI

Annual average NDVI (range -1 to 1)

GRAVHYY_03 - Growing season average NDVI

Growing season average NDVI (range -1 to 1)

SUPPORT CONTACT

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DATA SOURCE CONTACT

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